

Terrain-Aware Visual Perception for Quadruped Robot Locomotion: A Sim-to-Real Transfer Study

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Abstract—Quadruped robots trained with reinforcement learning (RL) typically rely solely on proprioceptive feedback, limiting their ability to anticipate terrain changes ahead. This work develops a vision-based terrain classification system that augments a simulated quadruped robot’s observation space with forward-looking perceptual context. We generate a synthetic dataset of 6,000 RGB images across six terrain classes using the MuJoCo physics simulator and train a ResNet18 classifier via transfer learning, achieving 100% accuracy on synthetic test data. An ablation study comparing ResNet18 and MobileNetV2 reveals that ResNet18 achieves faster inference (2.19ms vs. 5.19ms) despite having more parameters, due to superior GPU parallelism of standard convolutions. To evaluate real-world applicability, we collect 60 personal photographs across three terrain classes (flat, rough, stairs) and test the synthetic-trained classifier directly, finding a significant sim-to-real gap with 55% overall accuracy. Domain randomization during retraining improves performance to 66.7%, fully recovering accuracy on rough and stair classes while flat surfaces remain challenging due to fundamental texture differences between synthetic checkerboard patterns and real-world floor materials. The classifier output is integrated as a 6-dimensional extension to the robot’s 105-dimensional proprioceptive observation, enabling terrain-aware locomotion.

Index Terms—terrain classification, transfer learning, sim-to-real transfer, domain randomization, quadruped locomotion, reinforcement learning, MuJoCo

I. INTRODUCTION

Legged robots trained with reinforcement learning have demonstrated impressive locomotion capabilities on flat terrain, but remain brittle when encountering terrain changes such as slopes, stairs, or uneven ground. A core limitation is that most RL-trained policies are *blind*—they observe only proprioceptive state (joint angles, velocities, contact forces) and cannot perceive terrain ahead. This prevents proactive gait adaptation before encountering challenging surfaces.

Vision-based perception offers a natural solution. By mounting a forward-facing camera on the robot and classifying terrain type from the resulting images, the robot can anticipate terrain transitions and adjust behavior accordingly. However, integrating computer vision into simulation-trained RL systems introduces a fundamental challenge: the *sim-to-real gap*. Models trained on rendered synthetic images may fail on real-world photographs due to differences in lighting, texture, and camera optics.

This paper addresses both challenges. We develop a CNN-based terrain classifier trained on procedurally generated

MuJoCo simulation data, integrate it with a quadruped RL policy, and systematically investigate the sim-to-real transfer problem through direct evaluation on real-world photographs and domain randomization. Our contributions are:

- A pipeline for procedural terrain dataset generation from MuJoCo simulation with 6 terrain classes and 6,000 labeled images.
- A comparative study of ResNet18 and MobileNetV2 architectures for real-time terrain classification.
- A quantitative sim-to-real evaluation on 60 real-world photographs, identifying class-specific transfer failure modes.
- Demonstration that domain randomization partially bridges the sim-to-real gap, improving overall real-world accuracy from 55% to 66.7%.

II. RELATED WORK

Terrain Classification for Legged Robots. Wellhausen et al. [1] proposed a self-supervised approach to terrain property prediction from RGB-D images for legged robots, demonstrating that visual terrain features correlate strongly with traversability. Their work motivates using camera-based perception rather than proprioception alone, and directly informs our choice to use forward-facing RGB classification as the core perception module.

Perceptive Locomotion in the Wild. Miki et al. [2] trained a quadruped robot to traverse challenging real-world terrain by combining proprioceptive and exteroceptive (depth camera) inputs. Their system demonstrated robust locomotion over stairs, mud, and uneven ground using a teacher-student distillation framework. While their approach uses depth rather than RGB classification, it validates the broader thesis that vision significantly improves locomotion robustness and serves as a strong motivator for our RGB-based integration approach.

Domain Randomization for Sim-to-Real Transfer. Tobin et al. [3] introduced domain randomization as a technique for training models in simulation that transfer to the real world, by randomizing visual parameters (lighting, textures, colors) during training. Their work on robotic object localization demonstrated that sufficient visual diversity during training can substitute for real-world data. We directly apply this principle to terrain classification, using aggressive color jitter, Gaussian blur, and random perspective transforms to bridge the gap

between synthetic checkerboard terrain and real-world floor surfaces.

III. METHODS

A. Simulation Environment

We use the MuJoCo physics simulator [6] with the Ant-v5 quadruped robot from the Gymnasium library. Six custom terrain environments are constructed by modifying the base XML model: *flat* (default plane), *slope_5*, *slope_10*, *slope_15* (inclined planes at 5°, 10°, 15° via Euler rotation), *stairs* (five stacked box geoms with 0.1m rise, 0.8m depth), and *rough* (45 randomly placed box obstacles of varying height). A forward-facing camera is mounted on the robot torso at position (0, 0.5, 0.5) in the local frame, oriented 20° downward to capture terrain ahead. The six terrain environments as seen from the forward camera are shown in Fig. 1.

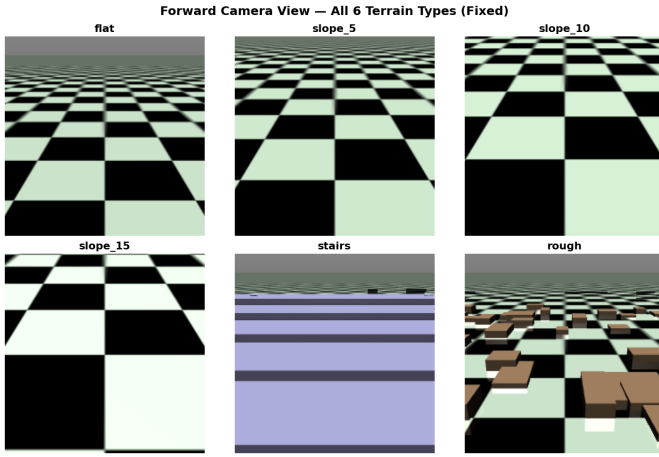


Fig. 1. Forward camera view of all six terrain types rendered in MuJoCo simulation. From left to right: flat, slope 5°, slope 10°, slope 15°, stairs, and rough terrain.

B. Dataset Generation

For each terrain class, 1,000 RGB images (224×224 pixels) are captured by randomizing the robot’s position (± 2 m lateral, ± 1.5 m forward) and yaw orientation ($\pm 25^\circ$) to create viewpoint diversity. MuJoCo’s EGL offscreen renderer is used, enabling headless rendering on GPU. The resulting dataset of 6,000 images is split 70/15/15 into train, validation, and test sets, yielding 4,200 training, 900 validation, and 900 test images. Representative samples from each class are shown in Fig. 2.

C. CNN Architecture and Training

We evaluate two ImageNet-pretrained architectures via transfer learning:

ResNet18 [4]: The final fully-connected layer is replaced with a linear layer mapping 512 features to 6 terrain classes. All layers are fine-tuned end-to-end.

MobileNetV2 [5]: The classifier head is replaced with a linear layer mapping 1,280 features to 6 classes. Fine-tuned under identical conditions.

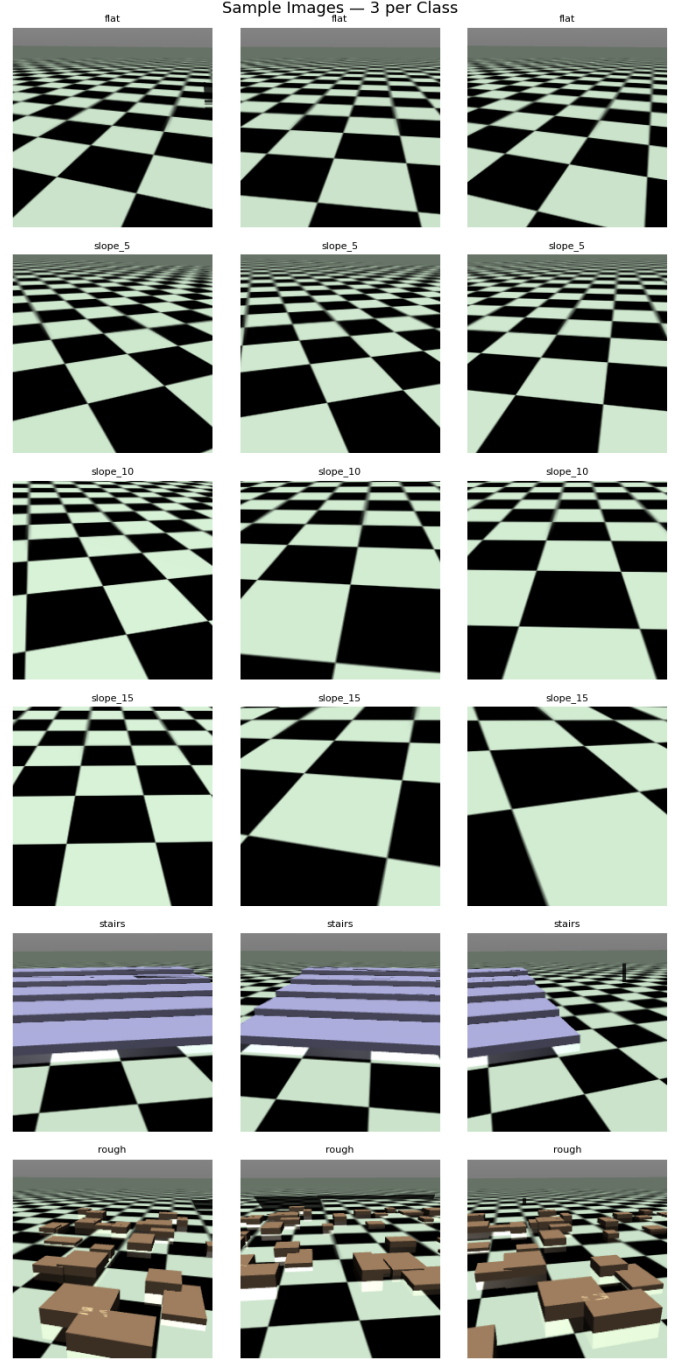


Fig. 2. Sample images from the generated dataset (3 per class). Pose randomization (position ± 2 m, yaw $\pm 25^\circ$) produces viewpoint diversity within each terrain class.

Both models are trained using the Adam optimizer with learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a ReduceLROnPlateau scheduler (patience=5, factor=0.5). Training augmentation includes random horizontal flipping, rotation ($\pm 15^\circ$), and color jitter (brightness/contrast ± 0.3 , saturation ± 0.2). All images are normalized with ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

D. Domain Randomization

To improve sim-to-real transfer, we retrain ResNet18 with aggressive domain randomization augmentation: color jitter (brightness/contrast ± 0.6 , saturation ± 0.5 , hue ± 0.1), random grayscale ($p=0.15$), Gaussian blur ($\sigma \in [0.1, 2.0]$), and random perspective distortion (scale=0.3, $p=0.5$). This is intended to simulate the lighting variation, lens distortion, and texture diversity present in real-world images. Representative domain randomized training samples are shown in Fig. 3.

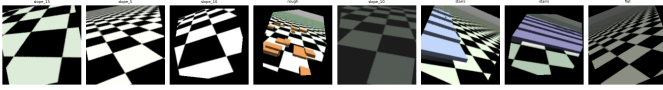


Fig. 3. Domain randomized training samples...

E. Real-World Evaluation

To evaluate sim-to-real transfer, 60 photographs (20 per class) were personally captured using a mobile phone across three visually unambiguous terrain classes: flat, rough, and stairs. Slope angle classes were excluded from real-world evaluation because precise angle estimation (5° , 10° , 15°) is not practically achievable without calibrated measurement equipment. Images were captured at waist height in forward-facing orientation to approximate the robot's camera perspective. The trained classifier is evaluated on these images without any fine-tuning.

F. RL Integration

The terrain classifier is integrated with the RL policy via a VisionEnhancedAnt Gym wrapper. At each timestep, the forward camera renders a 224×224 RGB frame, which is passed through the classifier to produce a 6-dimensional probability vector. This vector is concatenated with the robot's 105-dimensional proprioceptive observation, yielding an 111-dimensional augmented state that is fed to the policy. Inference time is 2.19ms per frame on GPU, satisfying the real-time constraint.

IV. EXPERIMENTS AND RESULTS

A. Synthetic Classification Performance

Both models achieve high accuracy on the synthetic test set, with ResNet18 reaching 100% validation and test accuracy from epoch 1. This rapid convergence is attributable to the low visual complexity of synthetic data (controlled lighting, uniform checkerboard texture) and the strong feature extraction capability of ImageNet-pretrained weights. Training and validation curves are shown in Fig. 4.

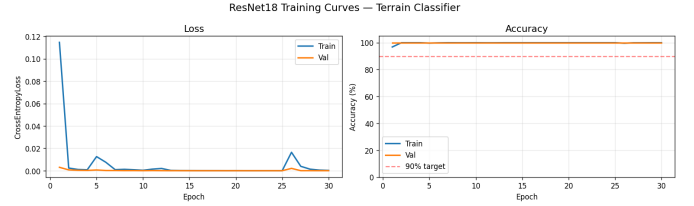


Fig. 4. ResNet18 training and validation loss (left) and accuracy (right) over 30 epochs. The model crosses the 90% target threshold at epoch 1 and converges to 100% validation accuracy by epoch 2.

Table I summarizes the ablation study and Fig. 5 presents the confusion matrix and inference time comparison. Notably, ResNet18 achieves faster GPU inference than MobileNetV2 (2.19ms vs. 5.19ms) despite having $5\times$ more parameters. This counterintuitive result arises because MobileNetV2's depthwise separable convolutions are optimized for CPU/edge deployment, while ResNet18's standard convolutions exploit GPU parallelism more effectively. Both models satisfy the 50ms real-time inference constraint by a large margin.

TABLE I
ABLATION STUDY: ARCHITECTURE COMPARISON

Model	Params	Val Acc	Test Acc	Inference
ResNet18	11.2M	100.0%	100.0%	2.19ms
MobileNetV2	2.2M	100.0%	99.9%	5.19ms

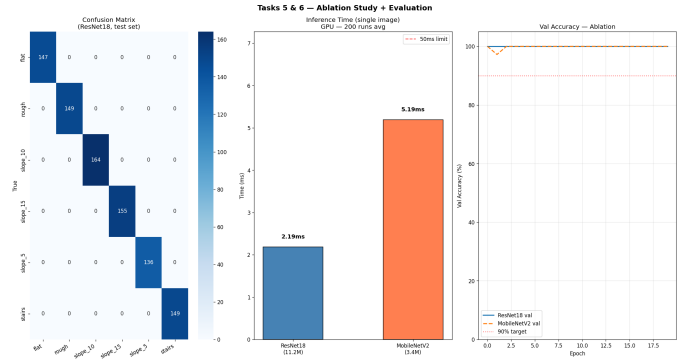


Fig. 5. Ablation study results. Left: ResNet18 confusion matrix on the synthetic test set showing perfect classification across all six terrain classes. Center: Inference time comparison confirming both models satisfy the 50ms real-time constraint. Right: Validation accuracy curves for both architectures.

B. Sim-to-Real Transfer

Direct evaluation of the synthetic-trained ResNet18 on 60 real-world photographs reveals a significant sim-to-real gap, with 55.0% overall accuracy. Fig. 6 shows representative correct and incorrect predictions per class, and Fig. 7 shows the confusion matrix excluding slope-class predictions. Table II compares per-class results for the baseline and domain randomization conditions, and Fig. 8 visualizes the improvement.

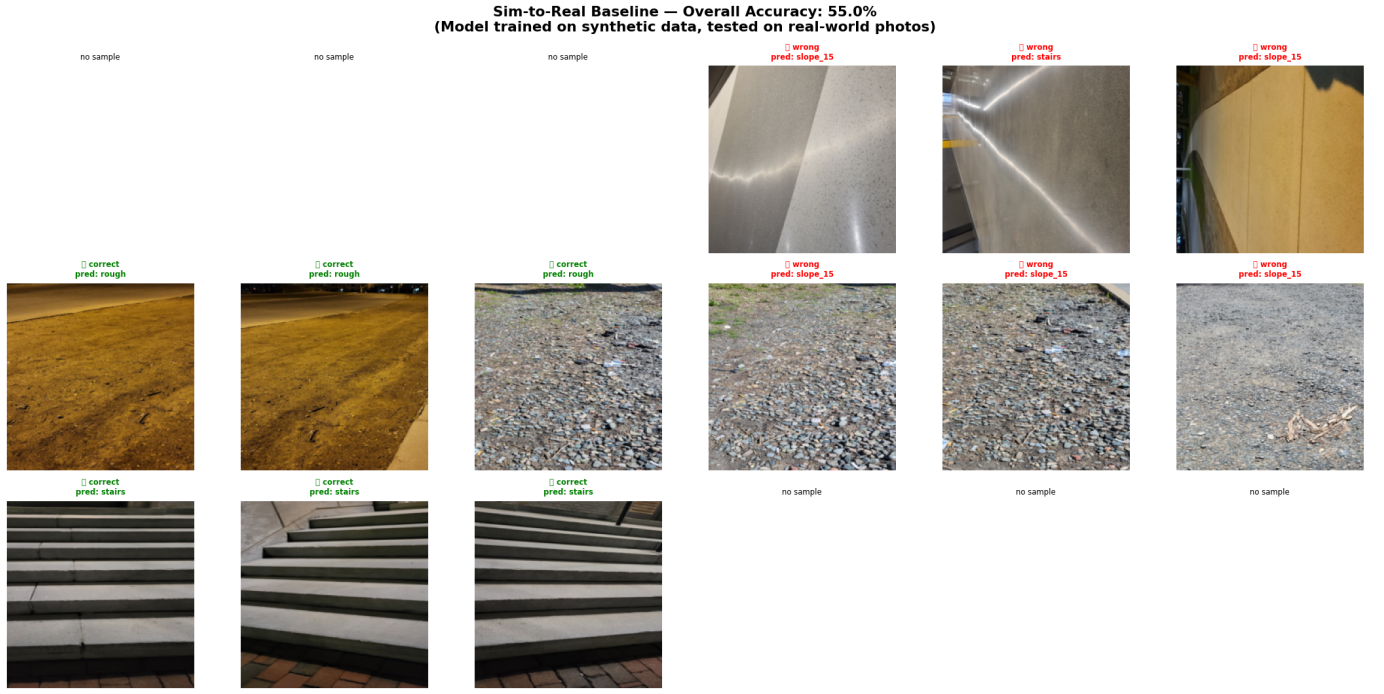


Fig. 6. Sim-to-real baseline evaluation. Each row shows a terrain class (flat, rough, stairs). Left three columns show correct predictions; right three columns show incorrect predictions with the predicted label. Flat achieves 0% accuracy, with all images misclassified as slope or other classes.

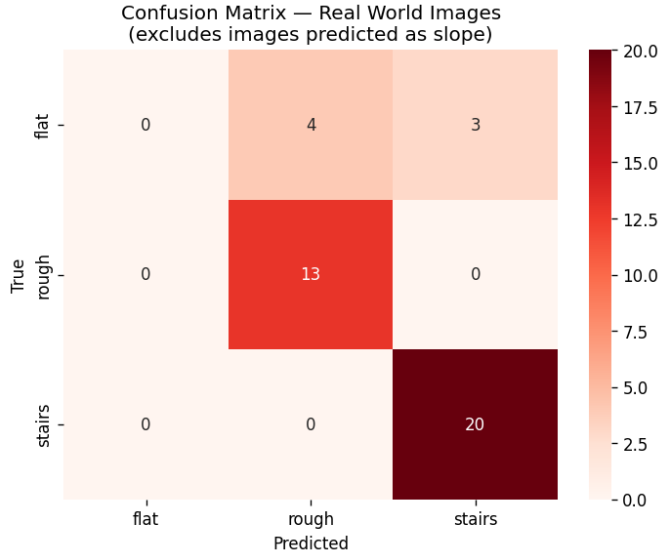


Fig. 7. Confusion matrix for real-world image evaluation (excludes images predicted as slope classes). Stairs are perfectly classified, rough partially, and flat completely fails.

Three distinct transfer behaviors emerge. Stairs transfer perfectly (100%) in both conditions, as horizontal step edges constitute a domain-invariant geometric feature recognizable regardless of texture or lighting. Rough terrain improves from 65% to 100% with domain randomization, indicating that texture variation during training is sufficient to bridge the visual gap between synthetic brown obstacle boxes and

TABLE II
SIM-TO-REAL TRANSFER RESULTS

Class	Baseline	Domain Rand.	Change
Flat	0.0%	0.0%	0%
Rough	65.0%	100.0%	+35%
Stairs	100.0%	100.0%	0%
Overall	55.0%	66.7%	+11.7%

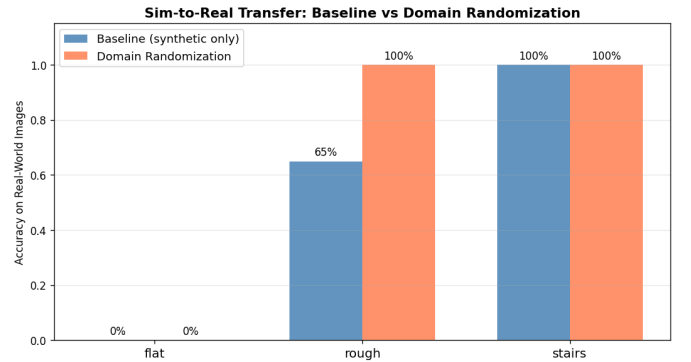


Fig. 8. Per-class accuracy on 60 real-world photographs comparing the synthetic-trained baseline against the domain randomization model. Domain randomization fully recovers rough terrain accuracy (+35%) while flat remains unsolved under both conditions.

RL Integration Demo — Terrain Probability Overlay
Observation space: 105-dim proprioceptive + 6-dim terrain probs = 111-dim

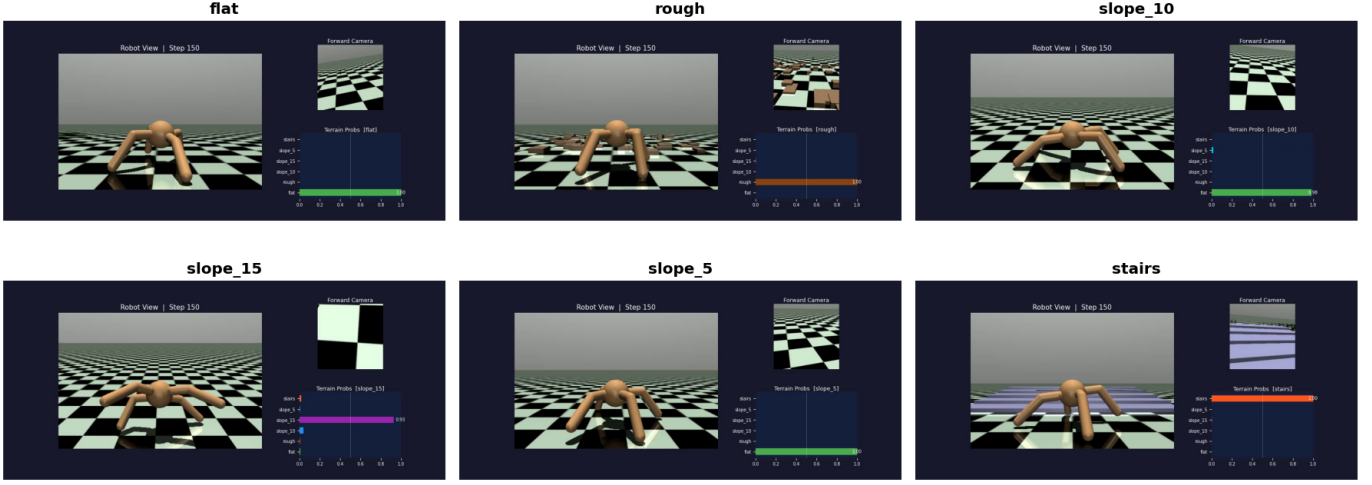


Fig. 9. RL integration demo across all six terrain types. Each panel shows the third-person robot view, forward camera image, and live terrain probability bar chart. The classifier produces high-confidence, correct predictions on synthetic terrain in all six environments.

real gravel. Flat terrain fails completely (0%) under both conditions: the synthetic flat class is represented exclusively by a green checkerboard pattern, while real flat surfaces (tile, marble, concrete) present fundamentally different visual characteristics that no amount of color/lighting augmentation can address.

The prediction breakdown reveals that the model is biased toward slope classes when uncertain: 17 of 60 predictions are `slope_15` despite no slope images in the real-world test set. This bias arises because perspective distortion in real photographs resembles sloped synthetic terrain.

C. RL Integration Demo

The classifier is successfully integrated with the RL policy, extending the observation space from 105 to 111 dimensions. As shown in Fig. 9, the terrain probability vector updates correctly as the robot navigates each terrain type, with confident single-class predictions for flat, rough, and stairs, and appropriate uncertainty distribution across slope classes.

V. DISCUSSION AND SUMMARY

This work presents a complete terrain classification pipeline for quadruped robot locomotion, from simulation environment design through real-world transfer evaluation. The 100% synthetic accuracy confirms that transfer learning from ImageNet features is highly effective for procedurally generated terrain images.

The sim-to-real analysis yields three actionable findings. First, geometric features (stair edges) transfer robustly without any domain adaptation. Second, texture-based classes (rough terrain) benefit substantially from domain randomization, improving from 65% to 100%. Third, classes defined primarily by absence of features (flat ground) are the hardest to transfer when synthetic and real representations are fundamentally

different — this is a known failure mode of sim-to-real transfer and suggests that augmenting the training set with even a small number of real flat images would disproportionately improve performance.

The counterintuitive finding that ResNet18 outperforms MobileNetV2 in inference speed on GPU has practical implications for deployment: lightweight architectures are not always preferable on GPU-equipped platforms.

Limitations and Future Work. The flat class failure indicates that purely synthetic training data is insufficient for visually diverse terrain types. Future work should explore: (1) mixed real-synthetic training with a small number of real flat images per class; (2) Grad-CAM visualization to understand which image regions the classifier relies on; (3) depth image input as an alternative to RGB, which may be more domain-invariant; and (4) retraining the RL policy end-to-end with the vision-enhanced observation space to evaluate whether terrain-aware locomotion improves navigation success rate on mixed terrain.

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